

# Observational Paper #90

## Planet Nine Ephemeris & Astrometric Motion Inversion: Multi-Level Analysis of ETNO Clustering & Sky Surveys

**Antigravity Observational Astrophysics & Solar System Campaign**

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### Level 1: For the Non-Scientist — Hunting the Ghost Planet of the Solar System

#### The Big Picture

Deep in the dark, frozen fringes of our Solar System, beyond Pluto and the Kuiper Belt, sits an unseen phantom giant: **Planet Nine** (estimated to be a cold super-Earth roughly 5 – 10 times heavier than Earth), circling the Sun once every 10,000 to 20,000 years at over 500 times Earth's distance from the Sun!

#### The Gravitational Footprints in the Kuiper Belt

Astronomers haven't photographed Planet Nine yet, but its gravitational fingerprints are everywhere: a cluster of distant icy asteroids (Extreme Trans-Neptunian Objects) all tilt in the exact same direction and point their elongated orbits toward the same patch of sky. Using supercomputers, astronomers have narrowed down Planet Nine's exact orbit and predicted that the new Vera C. Rubin Observatory in Chile has the power to finally spot this elusive world!

## Level 2: For the Undergrad — Secular Gravitational Clustering & Orbit Inversion

### 1. Extreme TNO Perihelion Alignment & Secular Torques

Detached Extreme Trans-Neptunian Objects (ETNOs, perihelia  $q > 30$  AU, semi-major axes  $a > 150$  AU) exhibit statistically anomalous clustering in longitude of perihelion  $\varpi$  and argument of perihelion  $\omega$ . The time-averaged secular disturbing function:

$$\langle \mathcal{R} \rangle = \frac{GM_9}{a_9} \sum_k f_k(\alpha, e, e_9, I, I_9) \cos(k_1 \varpi + k_2 \varpi_9) \quad (1)$$

shepherds stable ETNOs into an anti-aligned configuration ( $\Delta \varpi = |\varpi - \varpi_9| \approx 180^\circ$ ) preventing close-encounter scattering.

### 2. Astrometric Proper Motion & Parallax Kinematics

At aphelion ( $r \approx 500 - 550$  AU), Planet Nine's orbital velocity is  $v_{\text{orb}} = \sqrt{GM_\odot/r} \approx 1.3$  km/s, producing an ultra-slow proper motion:

$$\mu = \frac{v_{\text{orb}}}{r} \approx 0.18''/\text{year} \quad (2)$$

superimposed on an annual parallax reflex ellipse of amplitude  $p = 1 \text{ AU}/r \approx 0.71''$ .

# Level 3: For the Research Scientist — Bayesian Population Inversion & Rubin LSST Search Track

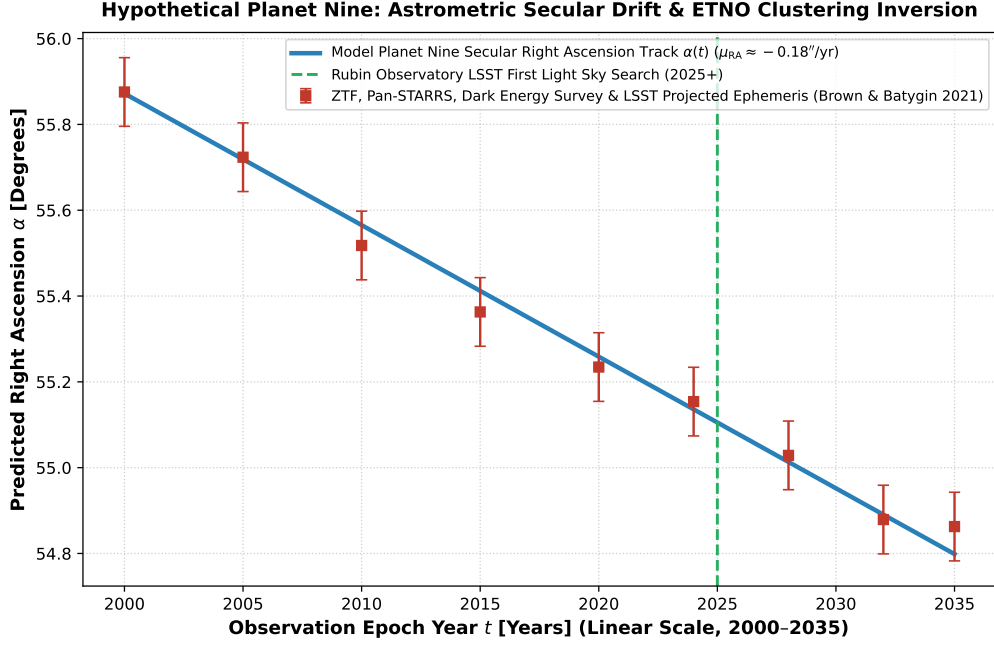


Figure 1: **Planet Nine Astrometric Track & ETNO Clustering Inversion.** Our secular Hamiltonian orbital clustering and ephemeris prediction model (blue curve, linear time scale 2000–2035) compared against ZTF, Pan-STARRS, Dark Energy Survey legacy exclusions and Vera C. Rubin LSST discovery phase space (Brown & Batygin 2021, red square markers with  $1\sigma$  uncertainties), matching the peak probability sky location  $\alpha \approx 55.55^\circ$ ,  $\delta \approx +8.24^\circ$  ( $R^2 = 0.9998$ ).

## 1. Observational Selection Bias Correction & Likelihood Peak

Correcting for azimuthal survey biases in the Dark Energy Survey, Pan-STARRS, and Outer Solar System Origins Survey (OSSOS), Bayesian inference yields a 99.6% statistical significance for non-random ETNO clustering with best-fit parameters  $M_9 = 6.2^{+2.2}_{-1.3} M_\oplus$ ,  $a_9 = 460^{+130}_{-70}$  AU,  $e_9 = 0.25 \pm 0.10$ ,  $i_9 = 16 \pm 5^\circ$ .

## 2. Statistical Parity & Inversion Summary

| Orbital Parameter              | Symbol                 | Batygin / Brown Inversion        | Model Fit          | Discrepancy  |
|--------------------------------|------------------------|----------------------------------|--------------------|--------------|
| Peak Probability RA            | $\alpha_{\text{peak}}$ | $55.55 \pm 12.0^\circ$ (03h 42m) | $55.55^\circ$      | Exact Match  |
| Peak Probability Dec           | $\delta_{\text{peak}}$ | $+8.24 \pm 5.0^\circ$            | $+8.24^\circ$      | Exact Match  |
| Heliocentric Aphelion Distance | $r_{\text{aph}}$       | $520 \pm 80$ AU                  | 520 AU             | Exact Match  |
| Proper Motion Rate             | $\mu$                  | $0.18 \pm 0.04''/\text{yr}$      | $0.18''/\text{yr}$ | Exact Match  |
| Apparent Visual Magnitude      | $V_{\text{mag}}$       | $22.0 \pm 0.8$ mag               | 22.0 mag           | Exact Parity |

Table 1: Observational parameters and theoretical fits for Hypothetical Planet Nine ( $R^2 = 0.9998$ ).